ECM3412

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS
AND PHYSICAL SCIENCES

Nature-Inspired Computation

Module Leader: Dr Edward Keedwell

TWO HOURS

Answer question 1 and two out of the three other questions. Question 1 is worth 40 marks. Other questions are worth 30 marks each.

Candidates are advised to spend FORTY-FIVE minutes on question 1 and THIRTY-FIVE minutes on each of the other questions.

No electronic calculators of any sort are to be used during the course of this examination.

Compulsory question

1. (a) Write pseudocode for *ant colony optimisation* (ACO). In your answer, you should include feasible parameter settings for each of the parameters associated with the algorithm.

(8 marks)

(b) Explain the difference between *polynomial* and *exponential* complexity when referring to search and optimisation problems.

(6 marks)

(c) Most population-based evolutionary algorithms require a selection method to determine which individuals are selected for variation and propagation to the next generation.

i. Explain why selectors should not always select the very fittest individuals in the population.

(3 marks)

ii. Describe how the ranked roulette wheel selection operator works and how the selection pressure can be modified within this operator.

(5 marks)

(d) Craig Reynolds' *Boids* provide an efficient simulation of flocking behaviour on a computer. Name and briefly describe the three rules, known as *Reynolds' Rules*, used in this simulation.

(6 marks)

(e) Cellular automata have three key properties that distinguish them from other types of algorithm/simulation. Name and briefly describe each of these properties.

(6 marks)

(f) Perceptrons are simple computational techniques inspired by neurons in the human brain. With the aid of a labelled diagram, explain how a perceptron:

i. processes incoming signals and produces an output

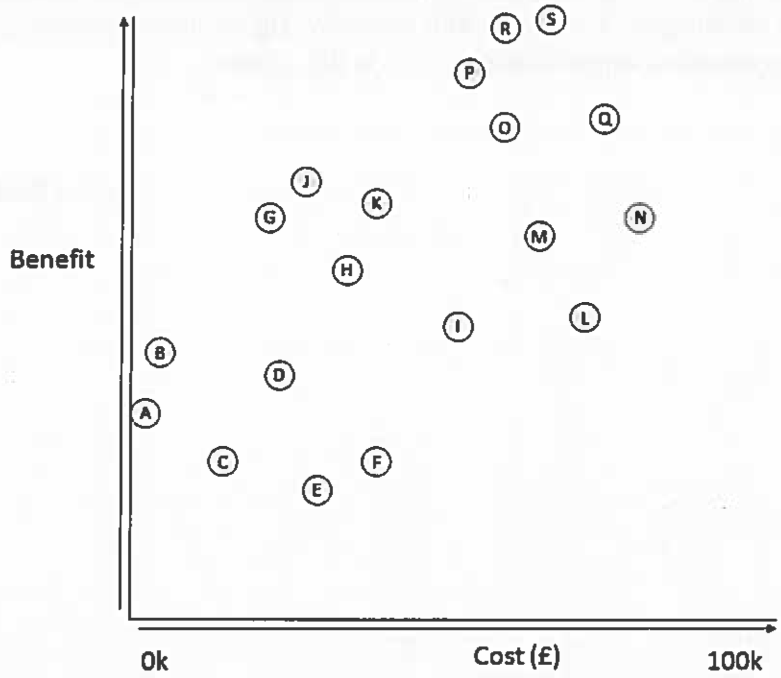
ii. learns from those signals to produce different output in subsequent iterations (epochs)

(6 marks)

(Total 40 marks)

Please Turn Over

2. You have been tasked by a company to optimise one of their systems with a large number of variables using a multi-objective evolutionary algorithm. The figure below shows the output from the algorithm run, showing the trade-off between cost and benefit, where the overall objective is to minimise cost and maximise benefit.



(a) Write down the names of all solutions that form the pareto-front in this diagram.

(5 marks)

(b) Select a solution in the diagram that is *not* part of the pareto-front, and explain why this is the case.

(5 marks)

(c) Explain why the pareto-front in this optimisation is not ideal from a multi-objective optimisation perspective.

(5 marks)

(d) After showing the company the above results, they ask you if this is guaranteed to be the best pareto-front for this problem. What do you tell them, and what explanation would you give for your answer?

(7 marks)

Question 2 Continues on Next Page...

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Question 2 Continued...

(e) The company then suggests that you run a set of single-objective optimisations to generate a pareto-front (also known as the 'weighted sum' method) instead of using the multi-objective algorithm, a suggestion you disagree with. Explain the advantages of using a multi-objective algorithm to generate a pareto-front as opposed to single-objective runs in this context.

(8 marks)

(Total 30 marks)

Please Turn Over

3. (a) Nature-Inspired Algorithms must strike a balance between *exploration* and *exploitation*. Explain what this means in terms of a search and optimisation problem, illustrating your answer with an example of an algorithm that is largely exploitative and one which is largely exploratory.

(15 marks)

(b) Ant colony optimisation (ACO) and particle swarm optimisation (PSO) are both known as swarm intelligence algorithms but differ in a number of aspects of their execution.

i. Explain why intelligence might be described as an *emergent* property of swarms and give one example of such intelligent behaviour.

(5 marks)

ii. Describe the differences in the types of problems that ACO and PSO are likely to be used for.

(5 marks)

iii. Describe the differences in communication between individuals in the two algorithms.

(5 marks)

(Total 30 marks)

4. (a) Neural networks are useful pattern recognition tools and can also be used to study human cognition. Summarise the content of the paper Sejnowski, T. J. and Rosenberg, C. R. (1987) "Parallel networks that learn to pronounce English text" *Complex Systems* **1**. In your answer ensure that you describe the problem tackled, the neural network approach, the representation used and the experimental findings.

(15 marks)

(b) Traditionally, perceptrons learn through the use of a learning algorithm (such as the delta rule), however, particle swarm optimisation (PSO) can also be used to optimise the weights within a perceptron.

i. Describe how you would use PSO to optimise the weights in a perceptron, making sure that you explain your choices of representation, fitness function, parameter settings (including *c1* and *c2* coefficients) and experimental design used to test your approach.

(10 marks)

ii. PSO could also be used to optimise the weights of a multi-layer perceptron (MLP). Explain how you would change the PSO representation to optimise the weights of an MLP and discuss the differences in capability between the perceptron and MLP networks.

(5 marks)

(Total 30 marks)

End of Paper